

Does the average BMI differ significantly among high school students across grade levels ?

Variables Definition

- >>> X variable : grade
- >>> Y variable : BMI

Statistical Method

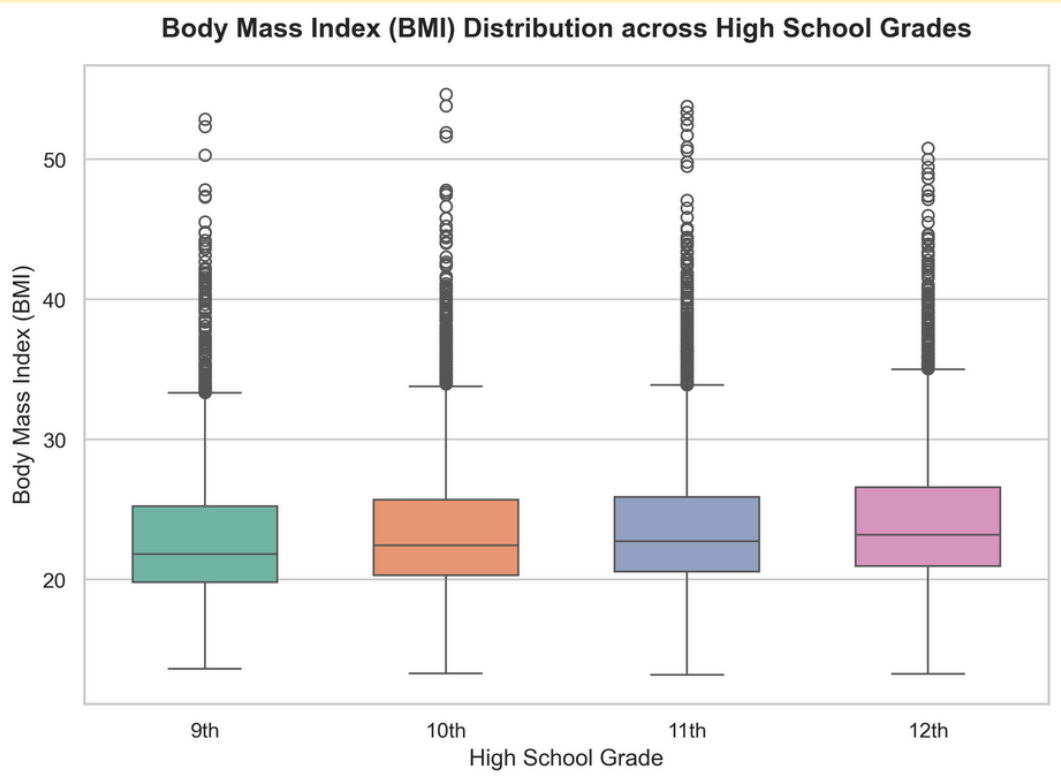
One-Way ANOVA

used to compare the means of the four independent groups and determine whether any of those means are statistically significantly different from one another.

Cleaning Data

grade/BMI	count	mean	min	max	std
9th	3162	23.125737	13.638946	52.882396	4.870024
10th	3248	23.591383	13.299383	54.655361	4.912784
11th	3273	23.995681	13.208280	53.776958	5.084975
12th	3351	24.397298	13.253470	50.792798	4.994888

EDA(Boxplot)



- >>> All four grade levels show highly similar BMI distributions and medians.
- >>> The visual difference is subtle, so the need for a formal ANOVA test to determine statistical significance.

Statistical Results

Hypotheses

- Null Hypothesis (H0): $\mu_{9th} = \mu_{10th} = \mu_{11th} = \mu_{12th}$ (Mean BMI is identical across all grades)
- Alternative Hypothesis (H1): At least one grade level has a different mean BMI.

Conclusion

- >>> Although the visual difference in the boxplot is highly subtle, the descriptive statistics reveal a steady upward trend: 9th-grade mean BMI is ≈ 22.2 , while 12th-grade mean BMI reaches ≈ 23.1 .
- >>> Driven by a massive sample size ($N > 10,000$), our ANOVA model has immense statistical power, proving this minor ≈ 1.0 BMI increase is statistically highly significant rather than random noise.
- >>> Real-World Impact: High school students experience a slight but genuine growth in body mass over time, which is heavily associated with natural biological development and adolescent lifestyle shifts

ANOVA output

	sum_sq	df	F	PR(>F)
grade	2897.587212	3.0	39.142723	3.577401e-25
Residual	321520.484866	13030.0		

- >>> Calculated p-value: $3.58e-25$ ($p < 0.05$)
Highly significant, reject H0